Prompting Small LLMs for Turkish Financial Event Detection: Does Translation or In-Context Learning Help?

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Understanding Language Models, Final Project

Abstract

In this project I aim to test whether small open LLMs manage to (3 to 4B parameters) detect high impact events in Turkish financial news headlines better when the headline is translated into English, and when we perform in-context learning (the prompt contains five examples). On 850 headlines from two years of news that I have scraped and labelled as a native Turkish speaker, I try to make three comparisons: 1) Turkish input against English input from an external translation system or from the model itself 2) zero-shot against few-shot prompting 3) the small models against a keyword lexicon and three frontier LLMs. I found that neither translation nor examples helped consistently: both mainly shifted how often a model answered YES, while its ranking ability (AUPRC) didn't change much. Frontier models were closer to my labeling, and the best small model matched the lexicon on the full set and exceeded it on the random sample.

1 Introduction

News moves stock prices, and a system that trades on news first has to recognise which headlines report events large enough to matter. For English, this kind of financial text classification is well served by domain models such as FinBERT (Araci, 2019) and benchmarks such as the Financial PhraseBank (Malo et al., 2014). For Turkish, public resources are not very abundant and they do not perform well much. During my bachelor thesis, I worked on a similar project (didn't involve LLM's) and my literature review there produced no concrete results for the Turkish models. So for this project, I aimed to try out small instruction-tuned LLMs, as they can be prompted for a new task without training data and run on my laptop (M4 Macbook, unfortunately I didn't had access to RTX5060 laptop I mentioned in my proposal) at no cost.

I came up with two main questions while deciding on how to proceed with the Turkish text. The first is whether to translate it to English or not. Machine translating the input into English and classifying the translation is a solid and I believe more "guaranteed" approach in non-english classification (Artetxe et al., 2023). Etxaniz et al. (2024) showed that the same model solves multilingual benchmarks better when the input is first translated into English, either by an machine translation system or by the model itself. The second question is if in-context learning helps, and what the model learns from these examples. Min et al. (2022) found that the correctness of the answers shown in the examples doesn't matter much where random answers reduce only a little accuracy. Wei et al. (2023) goes into more detail and says that small models largely ignore the example answers and rely on what they learned during the pretraining process, while large models manage to follow them even when they contradict with it. Both studies use English benchmarks so a lower resource language like Turkish and a specific domain task are a good test of how far these findings extend, but my aim for this project is not to go on a detailed review on these papers but rather follow my own idea for the corpus I have created, and these are the questions and papers that I found interesting.

I proceed with a binary task: does a Turkish financial news headline report a *high-impact event*, something that would significantly move the share price of a listed company or the Turkish market? Here "significantly" is a little subjective since there is no price data and my labeling depends purely on my understanding of Turkish language but a rule of thumb is around %5 change throughout the day, also note that unlike US stock market Turkish stock market has daily limit of +- %10. I label 850 headlines drawn from two years of news (main corpus is even larger, details at the 2.2 Corpus section), and evaluate three open 3–4B-parameter models in several prompting conditions, with three frontier LLMs as a reference. After completing most of the

steps I did some literature review to also compare with my approaches, and got inspired by translation comparison of Etxaniz et al. (2024) and the example label manipulation of Min et al. (2022) on natively Turkish text. My three main research questions are as follows:

RQ1 Does classifying an English translation of a headline, produced by an external translation system or by the model itself, outperform classifying the Turkish original?

RQ2 Do five in-context examples outperform zero-shot prompting, and does it matter whether the example answers come from a human annotator, from the model itself, or are random?

RQ3 How closely do the models labels agree with my labels, compared with a keyword lexicon and with frontier LLMs?

I find that neither translation nor in-context examples help consistently. Both mostly change how often a model answer with YES rather than how well it separates high-impact from random or regular headlines, and correct example answers matter little, as Min et al. (2022) and Wei et al. (2023) would predict for models of these sizes. The two Claude models agree with my labels best and are much stricter in assigning something as high-impact, while Gemini labels about as many headlines high-impact as I do but less precisely.

2 Methods

2.1 Task

The input is a single Turkish news headline. The output is binary: high-impact or not - my labels also have high-impact positive and high-impact negative seperation too. I consider a headline high-impact when it reports a concrete, new event (not commentary, a forecast, a price report or a list) that a reasonable investor would expect to move the share price of a specific listed company noticeably, or the broad Turkish market within that day. Both directions count: a large contract or a credit upgrade is as high-impact as an arrest or mass firings. I use headlines only, so labels and predictions reflect what the headline itself states.

2.2 Corpus

I use the headlines for a subsection of database I have built this summer. I (it would be more honest to say Claude as it wrote %95 of script code) scraped it during late-august. There are more detailed fields as well, and I am planning to extend this project with this dataset to try to earn some money from it :)

Source	Rows	First date	Avg Title Len
ROTABORSA	85,088	2022-11-28	56
DUNYA	61,329	2023-01-01	60
BORSAGUNDEM	48,289	2024-07-08	50
EKONOMIM	17,281	2022-01-20	57
KAP	16,579	2022-01-01	72
UZMANPARA	11,462	2014-08-01	38
MYNET	4,733	2026-01-01	60
Total	244,761	2014-08-01	56

For the subsection for this project, I use three business news outlets with different styles: ROTABORSA (company-centred, often naming the ticker), DUNYA (a national business daily) and BORSAGUNDEM (market news and commentary), as these are the most "clean" headline news sources, while the rest of the sources also heavily depend on the main body.

From 161,869 headlines published between 1 January 2024 and 22 August 2026, I built a frozen snapshot by removing 1) repeated news such as daily lists of the most traded stocks; 2) titles shorter than 20 characters or four words 3) exact duplicates within or across news websites, keeping the earliest copy. The snapshot contains 103,902 headlines; 83,507 fall into the 24-month test window (23 August 2024 to 22 August 2026). Its SHA-256 hash and the row count after each filter are released with the code. This (mostly the original database part) section took over a week for me and was very exhausting, and for the code of it I depended heavily on Claude.

2.3 Test set and example pool

High-impact headlines are rare, so a purely random sample would contain too few positives for evaluation. The test set therefore combines three strata (Table 1):

- **Random** (730): one headline per calendar day of the test window, with the outlet rotating daily (ROTABORSA, DUNYA, BORSAGUNDEM, . . .). Every outlet thus covers the whole period, so outlet is not confounded with time.

- **Lexicon** (60): headlines matching a keyword lexicon (Section 2.9), 20 per outlet.

Stratum	ROTA	DUNYA	BG	Total	Pos.
Random	245	243	242	730	88
Lexicon	20	20	20	60	21
Burst	28	18	14	60	14
Test set	293	281	276	850	123
core set	108	103	89	300	62
Example pool	25	8	7	40	3

Table 1: Labelled headlines per stratum and outlet (ROTA = ROTABORSA, BG = BORSAGUNDEM) and the number labelled high-impact (Pos.). The core set is the subset on which the few-shot conditions were run.

- **Burst** (60): one headline from each of 60 randomly chosen days on which a single company appeared in at least four headlines. This source of likely positives is independent of both the lexicon and the LLMs, so neither system decided which positives entered the test set.

All 850 headlines were labelled by me. Labelling was done in two stages: first all 120 enriched headlines and a random sample of 180 of the 730 random-stratum headlines (the core set of 300), then the remaining 550. The zero-shot conditions are evaluated on all 850; the few-shot conditions, which need nine runs per model and language, were run on the core set, so all comparisons involving them use these 300 headlines. I also report results on the random stratum alone, because its class balance reflects the expected rate of high-impact headlines (around 12.1). For labeling, I asked Claude to generate me a script so I can use with a few keystrokes, which fastened significantly.

A separate example pool was drawn from 1 January to 22 August 2024, before the test window, so that no event can appear both as an in-context example and as a test item. The first draw of 20 headlines (6 random, 9 lexicon, 5 burst) contained only one positive, too few for the example sets, so a second draw of 20 lexicon and burst headlines was added, giving a pool of 40.

2.4 Annotation

I labelled all test-set and pool headlines myself, as a native Turkish speaker, following a one-page guideline (help of my Sabancı Economy Master student friend and Claude) released with the data. The guideline defines the task as above and lists typical positives (legal action against a firm or its executives, regulatory sanctions and trading halts, corporate distress, disasters and operational shocks, credit-rating changes, major corporate actions, macro-economic or political shocks) and typical negatives (routine disclosures, analyst target prices, market reports, lists, commentary, small events). When unsure, borderline cases were decided by the label depends on whether the headline itself reports an event large enough to move prices, not on its tone or on a price move it describes. For each positive I also recorded the more likely price direction (up or down).

2.5 Translation

Headlines were translated into English with OPUS-MT (Tiedemann and Thottingal, 2020), using the transformer-big Turkish–English model `opus-mt-tc-big-tr-en` with beam size 4. The smaller `opus-mt-tr-en` was tried first and dropped because it mistranslated company names and tickers (e.g. *Anadolu Efes* → *Asia Minor Efes*). Company names are a particular problem for this task: *Şok Marketler*, a supermarket chain, becomes *Shock Markets*, which reads like market turmoil. Therefore before translation I replace company names with placeholders (`ENT1`, `ENT2`, ...) and restore them afterwards. Names are found with an alias list of 113 listed companies taken from BIST100, and as the run of capitalised words directly before a ticker in parentheses. The model copies the placeholders unchanged and still translates the Turkish case suffix that follows them (*ENT1'den* → *from ENT1*). Names were masked in 283 headlines; no placeholder was lost. I checked a random sample of 50 translations for adequacy and found no significant/consistent issues.

Self-translation. Following Etxaniz et al. (2024), each local model also translates the headlines itself. The prompt is plain text with four Turkish–English example pairs, the first sentences of the FLORES-200 development set no longer than 100 characters (NLLB Team et al., 2022), in the format `Turkish: ... English: ...`; the model continues after `English:` with greedy decoding until the end of the line. As in the original study, names are not masked. The model then classifies its own translation. I run this condition zero-shot only.

2.6 Models and prompting

I evaluate three open instruction-tuned models of similar size from different families: Qwen2.5-3B-Instruct (Qwen Team, 2024), Llama-3.2-3B-Instruct (Grattafiori et al., 2024) and the text-only

Rule	Example headline	Label
A price move is not an event; label its cause, if stated	*PAHOL Hisseleri İkinci Günde de Tavan Yaptı* (limit-up for a second day)	0
Price levels and market reports, however dramatic	*Brent petrol son 4 yılın en düşük seviyesinden işlem görüyor*	0
"Announced" is not "changed": a routine disclosure is 0, an announced event is judged on its content	*Ford Otomotiv (FROTO) 2024 yılı temettü kararını açıkladı!* *Alarko Holding (ALARK) dev satın almayı açıkladı!*	0 1
Forecasts and expectations, also by rating agencies; rating *actions* count	*S&P Global: TCMB'den Aralık ayına kadar indirim beklemiyoruz*	0
Registration of an earlier decision, or a change of the authorised capital ceiling	*Galatasaray'ın (GSRAY) yüzde 400 oranında sermaye tavanı artırımı tescil edildi*	0
Scale relative to the company; "dev" or a large amount signals materiality	*Garanti Bankası'ndan (GARAN) 38 milyonluk satış* *Graintürk Holding (GRTRK) dev satın almayı açıkladı*	0 1
Scheduled statistics and flows (macro data, PMI, fund and broker flows)	*Yabancı yatırımcılar pay piyasasında net 253,3 milyon dolar alım yaptı*	0
Foreign events only with a plausible link to Turkish markets	*İngiltere'den Rusya'ya yönelik yeni yaptırım kararları*	0
Legal action, distress or a trustee at a listed firm	*Banvit'ten (BANVT) soruşturma ve kayyım açıklaması*	1

Table 2: Rules applied to recurring borderline headlines, with examples from the data.

```
You classify headlines from Turkish financial
news. A headline is high-impact if it reports a
concrete, new event that would noticeably move
the share price of a specific listed company or
the Turkish stock market, in either direction.
Commentary, forecasts, price reports, lists of
stocks, routine disclosures and small events
are not high-impact.

Is the headline high-impact? Answer with Yes or
No.

Examples:
Headline: <example 1>
Answer: Yes
. . .

Headline: <target headline>
```

Figure 1: Prompt template. The `Examples` block is absent in the zero-shot condition.

Gemma-3-4B-it (Gemma Team, 2025). All run locally on a laptop (Apple M4, 16 GB) with MLX, quantised to 4 bits (group size 64).

Every prompt is a single turn in the chat format (Figure 1). The instructions are always in English; only the language of the headlines changes between conditions, so that RQ1 compares input languages rather than instruction languages (though I would also like to check this in future if it changes anything). In the few-shot conditions five example headlines with their answers come before the target headline, at the same language as the target.

Scoring. I do not generate text myself. For each headline I read the model's probability distribution over its first answer token and compute $p = P(Yes)/(P(Yes) + P(No))$. The prediction is *high-impact* if $p > 0.5$; p itself is used for threshold free metrics. With in context examples in the prompt, the models tried to copy the examples' format and begin their reply with "Answer:" (Gemma did so with almost all its probability mass), so in few-shot conditions for the replys that is started with "Answer:" next token is scored. The mean probability mass on Yes and No is then 1.00 for Qwen and Gemma in every condition. For Llama it is 1.00 zero-shot but 0.65–0.99 with examples, where some mass goes to line breaks; its few-shot scores are therefore less certain.

2.7 Conditions

Each local model is run with two input languages (Turkish, English via external trasnlation) in four prompting conditions, plus a zero-shot run on its self-translation.

- **Zero-shot**: no examples.
- **Gold**: five examples with my labels as answers.
- **Self**: the same five examples with the model's own zero-shot answers for them.
- **Random**: the same five examples with answers drawn uniformly from {Yes, No}, following Min et al. (2022).

The example headlines are identical across gold, self and random, so the three conditions differ only in the answers shown. Each example set contains two headlines the human labelled high-impact and

three labelled not, in random order. Because performance is known to vary with the choice of examples (Zhao et al., 2021), I draw three example sets from the pool and combine them via majority vote for prediction and mean for p.

2.8 Frontier LLMs

As a reference for RQ3 I also collect zero-shot labels from three frontier models, Gemini Flash 3.8, Claude Opus 5.5 and Claude Sonnet 5.5. Each receives the same instructions (Figure 1) and the 850 test headlines in Turkish or in the external English translation, in separate sessions per language. To keep them from cheating, each session sees only a folder with the headlines, renumbered, and the instructions; it writes one Yes/No answer per headline. Without token probabilities, they are evaluated on F1 and κ only.

2.9 Keyword lexicon

As a baseline I use a lexicon of 68 Turkish word stem patterns in seven categories (legal, regulatory, distress, disaster, rating, corporate, macro-political), for example *tutukla-* (arrest), *konkordato* (restructuring) and *not(unu) düşür-* (rating downgrade). It was written before any labelling and matches 7.1% of headlines in the test window with the help of my aforementioned Econ master friend and Claude. A headline is predicted high-impact if any pattern matches; the number of matches is its score. The same lexicon defines the lexicon stratum, so its recall on the full test set is optimistic while recall on the random and burst strata is unaffected.

2.10 Evaluation and statistics

I report precision, recall and F1 for the high-impact class, Cohen's κ against the human labels, and two threshold-free metrics computed from p: the area under the ROC curve and the area under the precision recall curve (AUPRC), it is a nice choice when classes are imbalanced. 95% confidence intervals for F1 and κ come from 1,000 bootstrap resamples of test items.

Conditions are compared on the same items, so I use exact McNemar tests on per-item correctness:

- **RQ1**: Turkish vs. English for each model and prompting condition (12 tests), and, zero-shot, self-translated English vs. Turkish and vs. external MT (6 tests).
- **RQ2a**: zero-shot vs. gold for each model and language (6 tests).
- **RQ2b**: gold vs. random and gold vs. self for each model and language (12 tests).

p-values are corrected with the Holm procedure within each family. For RQ3 I compare each condition's κ with the human labels to that of the keyword lexicon and of the frontier LLMs. (The choices of the tests and their coding has been heavily done with the help of Claude)

2.11 Hypotheses

H1 English input yields higher accuracy than Turkish input (Etxaniz et al., 2024). Because my models are small, I expect self-translation to be not as good as external machine translation, as Etxaniz et al. (2024) found for their smallest models.

H2a Five gold examples improve over zero-shot prompting.

H2b Replacing gold answers with random ones costs little (Min et al., 2022), because models of this size largely ignore the example answers (Wei et al., 2023).

H3 The best local-model condition agrees with the human labels more than the keyword lexicon does, but not as much as the frontier LLMs do.

3 Results

Table 3 gives the zero-shot results on all 850 test headlines, Table 4 and Figure 2 all prompting conditions on the 300-headline core set. Accuracy is high for every system (62–87%) since most headlines are negative so there is a class imbalance; therefore I prefer F1, κ and AUPRC. All p-values below are Holm-corrected; the McNemar tests compare per-item correctness, so they mostly reflect how many negatives a condition gets wrong.

RQ1: Translation. English input did not help consistently throughout the experiment (Table 3). Gemma and Qwen scored about the same in all three input languages, with no significant differences. Llama is the exception: on Turkish it almost never answered *Yes* (14 of 850 headlines, F1 .09), on English it answered more often (118, F1 .33). Its AUPRC didn't change much (.29 vs. .31), so translation shifted Llama's decisions rather than its

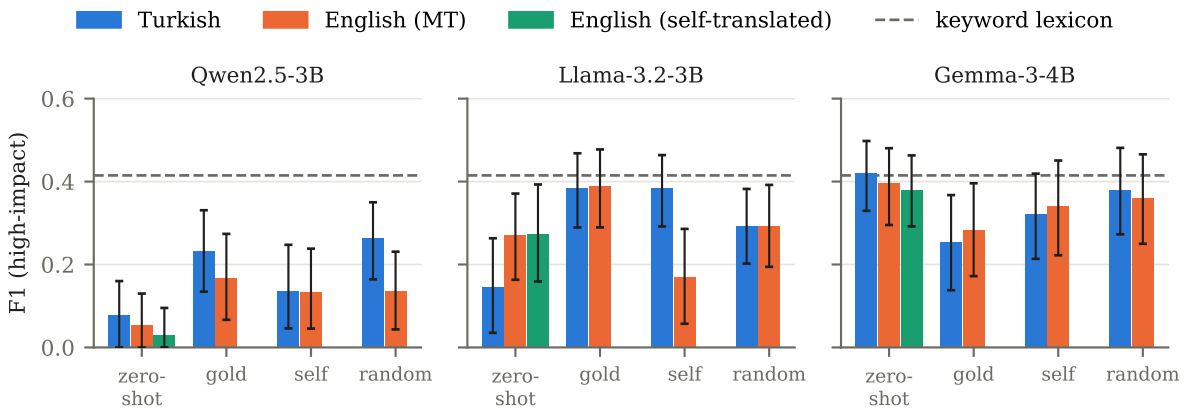


Figure 2: F1 for the high-impact class on the 300-headline core set, with 95% bootstrap intervals. The dashed line is the keyword lexicon. The examples change F1 more than the input language does, and in different directions for different models. Figure code is generated via Claude

System	Input	F1 [95% CI]	κ	AUPRC
Qwen2.5-3B	TR	.11 [.05, .18]	.03	.21
	EN (MT)	.10 [.04, .17]	.03	.19
	EN (self)	.03 [.00, .08]	.01	.19
Llama-3.2-3B	TR	.09 [.03, .15]	.06	.29
	EN (MT)	.33 [.25, .41]	.22	.31
	EN (self)	.30 [.23, .37]	.18	.27
Gemma-3-4B	TR	.36 [.30, .42]	.19	.26
	EN (MT)	.34 [.29, .40]	.18	.26
	EN (self)	.36 [.31, .42]	.20	.30
Claude Opus 5.5	TR	.34 [.24, .43]	.29	–
	EN (MT)	.36 [.27, .45]	.31	–
Claude Sonnet 5.5	TR	.34 [.25, .43]	.29	–
	EN (MT)	.35 [.26, .44]	.31	–
Gemini Flash 3.8	TR	.35 [.27, .43]	.25	–
	EN (MT)	.34 [.26, .42]	.22	–
Keyword lexicon	TR	.32 [.24, .40]	.22	.21

Table 3: Zero-shot results on all 850 labelled test headlines (123 high-impact). The frontier models give Yes/No answers only, so they have no AUPRC.

understanding. Self-translation wasn't significantly different from external translation. Both Claude models gave the same answers in Turkish and English on 98% of headlines. Gemini was the only frontier model affected by the language: it was more accurate on Turkish than on English (%82 vs. %79, $p = .026$).

RQ2: In-context examples. Examples changed the models' behaviour somewhat significantly, but mostly by shifting how often they answered *Yes* (Table 4, Figure 2). With gold examples Gemma became more conservative (which moves it away from LLM labeling to my labeling more) its accuracy and AUPRC increased (AUPRC .31 → .38), but its F1 dropped (.42 → .26) because it missed more events. Llama was different, from almost never to often answering *Yes*, which raised its F1 on Turkish (.14 → .39) but lowered its accuracy. Qwen stayed close to chance. Whether the example answers were correct mattered little: only two of twelve gold-vs-random and gold-vs-self comparisons were significant.

RQ3: Agreement with human labels. Agreement was low for every system (Table 3). The two Claude models agreed best (κ .29–.31), followed by Gemini (.25 Turkish .22 English) and Llama on English and the keyword lexicon (.22). The Claude models were much stricter than me: they labelled 4–6% of headlines high-impact, compared to my 14.5%, so their answers were mostly right but missed most events. On the random stratum, Llama on English (κ .26) was above the lexicon (.14) and close to Claude.

Direction. Every system found the my *up* events more often than the *down* events: zero-shot recall on up vs. down events was .67 vs. .44 for Gemma (English) and .31 vs. .19 for Claude (English).

4 Discussion

The most obvious pattern for me across all three questions is that the small models' decisions were depending less on what they understood about a headline than by how much they said *Yes*. Their ranking ability, measured by AUPRC, stayed within a narrow band across languages and prompts (.19–.38), while their *Yes* rates ranged from 1% to

		Zero-shot		Gold		Self		Random	
Model	Input	F1	AUPRC	F1	AUPRC	F1	AUPRC	F1	AUPRC
Qwen2.5-3B	TR	.08	.26	.23	.25	.14	.24	.26	.23
	EN (MT)	.05	.21	.17	.23	.13	.24	.13	.22
Llama-3.2-3B	TR	.14	.37	.39	.30	.38	.28	.29	.25
	EN (MT)	.27	.31	.39	.29	.17	.31	.29	.28
Gemma-3-4B	TR	.42	.31	.26	.38	.32	.36	.38	.36
	EN (MT)	.40	.32	.28	.38	.34	.35	.36	.33

Table 4: Prompting conditions on the 300 test headlines scored in every condition (62 high-impact). Few-shot predictions are the majority vote over three example sets; AUPRC uses the mean p.

43% which is a significant variance. Translation and examples mostly moved this rate, and whether that helped F1 or accuracy depended on where a model started.

RQ1. H1 is not supported. Etxaniz et al. (2024) found a consistent gain from English input; I didn't found for Gemma and Qwen, and for Llama a slight gain in F1. One explanation that I came up with is the origin of the text they used - their benchmarks were translated from English, so translating them back somewhat recovers the original but my headlines are native Turkish, and translating them adds errors instead (*dev satın alma*, "a giant acquisition", became "acquisition of giants"). Similar to their findings, self-translation was no better than external translation for small models. The Claude models' answers barely depended on the language, and Gemini was slightly better on Turkish, which shows that the language gap matters less than the model's understanding of the task, which is expected since Turkish understanding would be way improved for them compared to detecting these ups and downs.

RQ2. H2a holds only in part: five gold examples improved accuracy and AUPRC for Gemma, but lowered its F1; for Llama and Qwen they raised F1 but not accuracy or AUPRC. H2b is largely supported: in ten of twelve comparisons, replacing the gold answers with random or self-generated ones made no significant difference, consistent with Min et al. (2022). Where answers did matter, the models appear to follow the share of YES answers rather than which headlines were labelled YES, consistent with Wei et al. (2023)'s finding that small models rely on their prior knowledge rather than on the input context examples.

RQ3. H3 is half holding. On the random stratum the best local model agrees with the annotator clearly more than the lexicon does and about as much as the frontier models, but this isn't the case for the full test set where the lexicon is as good. Even the frontier models reached only $\kappa \approx .3$. Part of this is the difficulty of the task, where it is a judgement heavy task. The frontier models were conservative where I was not as I discussed.

Limitations. The gold labels came from only me and technically I am not an expert on finance. The input is the headline alone, which sometimes understates or hides the event. Machine translation adds its own errors but were mostly fixed by masking the corporate names. Instructions were always in English, so the Turkish condition mixes two languages in one prompt, which might have some negative effects on models - a future question. Finally, *high-impact* is defined by human judgement, not by measured price reactions.

Future work. There is a lot of future work to do, and I am planning to continue doing so individually. I want to use all my 250k data on a better compute. I also want to include "stock price data" which would be the best label to have as ground truth. I want to perform more experimentation (like the aforementioned mixing instruction language etc.) - because I don't think Turkish language performance on LLM's (and specifically finance domain) is researched in depth. I am also currently taking the LLM Engineering course and have enjoyed doing this project, so if I can get good enough with understanding LLM's I would love to be able to come up with proper and detailed experimentations to contribute to this field.

5 Conclusion

I tried to answer whether small open source LLMs label high-impact events in Turkish financial headlines better when the headline is translated into

English, and when they are given examples. On 850 headlines labelled by me as a native speaker, neither helped consistently. Translation and examples mainly changed how often the models answered YES, not how well they ranked headlines, and whether the example answers were correct didn't affect much. The two Claude models agreed with my labels best but were much stricter than me, Gemini was as less strict as me but less precise. For Turkish financial news, the practical lesson is to calibrate a small model's decision threshold before worrying about translation or prompt examples.

Data and code availability

The source code is available at https://github.com/ahmetalperengungor/UT_UnderstandingLM_Course_Project.git.

AI usage declaration

Throughout the coding part Claude has been used extensively via the VSCODE extension. However, except for some parts of the statistical tests (choosing which to use and conduct them), they weren't involved with ideas and brainstorming. The below parts of the code are almost fully by Claude: 1) Code for statistical tests and figures 2) A helper script for me to label quickly 3) Conducting subagents and deploying & prompting open source models. 4) Web scraping of my original dataset. Rest of the coding followed standard Claude code procedures, writing together, refining and changing code, reviewing etc. Except for the coding, Claude was used for creating the Latex code from my Word report. For writing the report, they are used only in some small parts (tests), they do not take part in writing large chunks of text, or writing in any part of the argumentative parts. The project idea, research questions, hypothesis and overall pipeline is fully my idea and I am planning to continue working on it.

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